Lecture 2: Interest and Time

MNY101 Money Basics

Sample course for trying Smazy

Why money has a time value

- Money today can be saved and earn interest
- So 100 today is worth more than 100 next year
- This idea builds on inflation from Lecture 1

Simple interest

- Interest is paid only on the amount first saved
- Formula: interest = principal x rate x years
  - Example: 1,000 at 5% for 3 years earns 150

Compound interest

- Interest is paid on the principal and on earlier interest
- Formula: future value = principal x (1 + rate) ^ years
  - Example: 1,000 at 5% for 3 years grows to 1,157.63
  - The extra 7.63 is interest earned on interest

The rule of 72

- A quick way to estimate how long money takes to double
- Years to double = 72 / interest rate in percent
  - At 6%, money doubles in about 12 years
  - At 9%, money doubles in about 8 years

Present value

- Present value is what a future amount is worth today
- Formula: present value = future amount / (1 + rate) ^ years
  - Example: 1,157.63 in 3 years at 5% is worth 1,000 today
- Homework 1 practises these calculations